WORKED SOLUTION

STEP III 1992, Question 9

Pure • Polar Equation of Ellipse

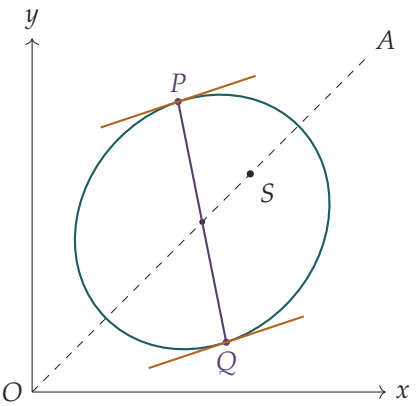
STEP III 1992, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q9

The key idea

Polar coordinates are measured from S with the initial line pointing along $y = x$, so a point at polar angle θ has Cartesian direction $\theta + \frac{\pi}{4}$ from S. Writing x and y as functions of θ and differentiating gives the gradient. For the second part, the property of a diameter we can actually use is that an ellipse is symmetric about its centre: the tangents at the two ends of a diameter are *parallel*. So we set the two gradients equal and let half-angle trigonometry do the rest.



In the sketch ($e = \frac{1}{2}$), PQ is a diameter through the centre, and the tangents at P and Q are parallel.

Solution

The gradient of the tangent

Let S be the point (s, s) with $s > 0$, since S lies on the line $y = x$ in the first quadrant. A unit vector along SA is $\frac{1}{\sqrt{2}}(1, 1)$, at angle $\frac{\pi}{4}$ to the x-axis. So the point with polar

coordinates (r, θ) about S is

$$x = s + r\cos\phi, \qquad y = s + r\sin\phi, \qquad \text{where } \phi = \theta + \tfrac{\pi}{4}.$$

On E we have $r = \ell/(1 + e\cos\theta)$, so

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{\ell e\sin\theta}{(1+e\cos\theta)^2} = \frac{r\,e\sin\theta}{1+e\cos\theta}.$$

Writing r' for $\mathrm{d}r/\mathrm{d}\theta$, and using $\mathrm{d}\phi/\mathrm{d}\theta = 1$,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = r'\cos\phi - r\sin\phi, \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = r'\sin\phi + r\cos\phi.$$

Only the ratio matters, so we multiply both by the positive quantity $(1 + e\cos\theta)/r$:

$$\frac{1+e\cos\theta}{r}\frac{\mathrm{d}x}{\mathrm{d}\theta} = e\sin\theta\cos\phi - (1+e\cos\theta)\sin\phi = e\sin(\theta-\phi) - \sin\phi = -\tfrac{e}{\sqrt{2}} - \sin\phi,$$
$$\frac{1+e\cos\theta}{r}\frac{\mathrm{d}y}{\mathrm{d}\theta} = e\sin\theta\sin\phi + (1+e\cos\theta)\cos\phi = e\cos(\theta-\phi) + \cos\phi = \tfrac{e}{\sqrt{2}} + \cos\phi,$$

since $\theta - \phi = -\frac{\pi}{4}$. Now $\sin\phi = \frac{1}{\sqrt{2}}(\sin\theta + \cos\theta)$ and $\cos\phi = \frac{1}{\sqrt{2}}(\cos\theta - \sin\theta)$. Multiplying through by $\sqrt{2}$, a tangent vector at $\theta = \alpha$ is

$$\mathbf{d}(\alpha) = (-(\sin\alpha + \cos\alpha + e),\ \cos\alpha - \sin\alpha + e). \qquad (*)$$

Hence

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\cos\alpha - \sin\alpha + e}{-(\sin\alpha + \cos\alpha + e)} = \frac{\sin\alpha - \cos\alpha - e}{\sin\alpha + \cos\alpha + e},$$

as required. (Where $\sin\alpha + \cos\alpha + e = 0$ the tangent is vertical; $(*)$ still holds there.)

Ends of a diameter

Ask yourself How can we turn "PQ passes through the centre" into an equation in α and β, when we haven't found the centre?

We use the first part. An ellipse is symmetric under a half-turn about its centre C. If P and Q are the ends of a diameter, the half-turn swaps them and maps the tangent at P to the tangent at Q. A half-turn sends every line to a parallel line, so *the tangents at P and Q are parallel*.

So, with $\theta = \alpha$ at P and $\theta = \beta$ at Q, the vectors $\mathbf{d}(\alpha)$ and $\mathbf{d}(\beta)$ from $(*)$ are parallel. Working with vectors rather than gradients means we never divide by zero when a tangent is vertical. Write $s_\alpha = \sin\alpha$, $c_\alpha = \cos\alpha$, and so on. The condition is

$$(s_\alpha + c_\alpha + e)(c_\beta - s_\beta + e) = (s_\beta + c_\beta + e)(c_\alpha - s_\alpha + e).$$

Expanding, the terms $c_\alpha c_\beta$, $-s_\alpha s_\beta$, ec_α, ec_β and e^2 appear on both sides and cancel, leaving

$$s_\alpha c_\beta - c_\alpha s_\beta + es_\alpha - es_\beta = s_\beta c_\alpha - c_\beta s_\alpha + es_\beta - es_\alpha,$$

that is,

$$\sin(\alpha-\beta)+e(\sin\alpha-\sin\beta)=0.$$

Now use $\sin(\alpha-\beta)=2\sin\frac{\alpha-\beta}{2}\cos\frac{\alpha-\beta}{2}$ and $\sin\alpha-\sin\beta=2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$:

$$2\sin\tfrac{\alpha-\beta}{2}\left(\cos\tfrac{\alpha-\beta}{2}+e\cos\tfrac{\alpha+\beta}{2}\right)=0.$$

? **Ask yourself** Can we cancel $\sin\frac{\alpha-\beta}{2}$?

Yes. P and Q are different points, so α and β don't differ by a multiple of 2π, and $\sin\frac{\alpha-\beta}{2}\neq 0$. Hence

$$\cos\tfrac{\alpha-\beta}{2}+e\cos\tfrac{\alpha+\beta}{2}=0.$$

Expanding with $\frac{\alpha}{2}$ and $\frac{\beta}{2}$:

$$(1+e)\cos\tfrac{\alpha}{2}\cos\tfrac{\beta}{2}+(1-e)\sin\tfrac{\alpha}{2}\sin\tfrac{\beta}{2}=0.$$

Dividing by $(1-e)\cos\frac{\alpha}{2}\cos\frac{\beta}{2}$ (here $1-e>0$ because E is an ellipse) gives

$$\tan\tfrac{\alpha}{2}\tan\tfrac{\beta}{2}=-\frac{1+e}{1-e},$$

as required. The division is valid unless $\cos\frac{\alpha}{2}$ or $\cos\frac{\beta}{2}$ is zero. If, say, $\alpha=\pi$, the equation forces $\sin\frac{\beta}{2}=0$, so $\beta=0$. That diameter is the major axis along OSA, and there the tangent product isn't defined. Every other diameter satisfies the result.

Checks

- At $\alpha=0$ (the vertex nearer S) the formula gives gradient $\frac{-1-e}{1+e}=-1$. That's right: the tangent at the end of the major axis is perpendicular to $y=x$.
- With $e=\frac{1}{2}$ the product should be -3. Take $\alpha=\frac{\pi}{2}$, so $\tan\frac{\alpha}{2}=1$. Then $\tan\frac{\beta}{2}=-3$, so $\sin\beta=-\frac{3}{5}$ and $\cos\beta=-\frac{4}{5}$. From $(*)$, $\mathbf{d}(\alpha)=(-\frac{3}{2},-\frac{1}{2})$ and $\mathbf{d}(\beta)=(\frac{9}{10},\frac{3}{10})$, which are parallel. ✓
- Numerically, for 2000 random choices of ℓ, e, S and α, the gradient formula agreed with a finite-difference derivative, and reflecting P in the centre (at distance $\ell e/(1-e^2)$ from S along AS) gave a point Q on E whose β satisfied the identity to 10^{-13}.

Common slips

- Forgetting the $\frac{\pi}{4}$ rotation and using $x=r\cos\theta$, which gives the gradient in the wrong frame.
- Dividing by $\sin\frac{\alpha-\beta}{2}$ without saying why it is non-zero.
- Trying to find the centre and impose "C is the midpoint of PQ" directly. It works but is far messier than using parallel tangents.

Beyond STEP

Parallel tangents at the ends of a diameter are part of the theory of *conjugate diameters*: the midpoints of a family of parallel chords lie on a diameter, and the tangents at its ends are parallel to those chords. Under an affine map that turns the ellipse into a circle, all of this becomes elementary circle geometry, which is one reason affine and projective geometry are so powerful. The half-angle tangents here are also the standard parametrisation used for orbits in celestial mechanics. For a good first-year route into this geometry, try Needham, *Visual Complex Analysis*, or Siklos, *Advanced Problems in Mathematics*.

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